A DATA VISUALIZATION PROJECT

GLOBAL AI ADOPTION IN EDUCATION

Tableau Dashboard, Data Story and Flask Web Integration

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1. Introduction

Artificial Intelligence (AI) tools such as ChatGPT, Google Gemini, Microsoft Copilot and Khanmigo are rapidly changing the way students learn and teachers teach. From instant doubt-solving and personalised practice to lesson planning and content creation, AI is becoming a regular part of education systems around the world.

However, the pace of adoption is not the same everywhere. It depends on factors such as internet access, the education level of a country, the difference between urban and rural areas, and the availability of AI tools in schools. To understand this changing picture, data needs to be explored visually rather than read as rows of numbers.

This project, **Global AI Adoption in Education**, analyses a dataset on AI usage in education across ten countries from 2015 to 2026. Using Tableau, the data has been converted into six visualizations, combined into an interactive dashboard, and narrated through a six-scene data story. The published Tableau views have then been embedded into a Flask web application so that they can be accessed from a single web page.

2. Problem Statement

Information about AI usage in education is usually scattered across surveys and reports, and is often presented in tables that are difficult to compare. Decision makers, teachers and students cannot easily answer simple questions such as: *How fast is AI adoption growing? Which countries use AI the most?*

Do students use AI more than teachers? Is rural usage lower than urban usage? Which AI tool is the most popular?

There is a need for a clear, interactive and visual way of presenting this information. This project addresses the problem by building a Tableau-based dashboard and story, and by making them accessible through a simple web application.

3. Objectives

The main objectives of this project are:

- To study the growth of AI adoption in education over the years 2015 to 2026.
- To compare AI adoption across different countries.
- To compare AI usage among students and teachers, and between rural and urban areas.
- To identify which AI tools are most commonly used for educational purposes.
- To design six clear Tableau visualizations and combine them into one interactive dashboard.
- To build a Tableau Story that presents the findings in a logical sequence.
- To integrate the published Tableau Dashboard and Story into a Flask web application.

4. Methodology / Project Workflow

The project follows the workflow shown in the diagram below. Each step feeds into the next, starting from the raw CSV dataset and ending with a web application that displays the final Tableau work.

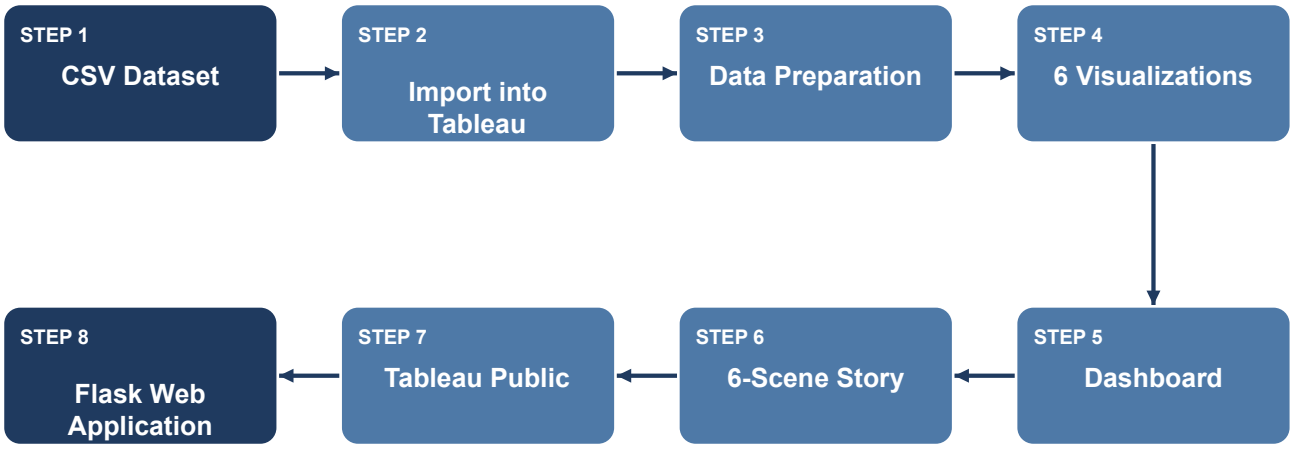


Figure 0: Project workflow

Explanation of the workflow

Step 1 - CSV Dataset. The project begins with a CSV file named *Global AI in Education*. It stores country-wise and year-wise values for AI usage by students and teachers, school adoption, internet penetration, education index and the most popular AI tool.

Step 2 - Import into Tableau. The CSV file is connected to Tableau as a data source. Tableau automatically identifies fields as dimensions (Country, Year, Top AI Tool) or measures (the percentage and index values).

Step 3 - Data Preparation. The fields are checked for correct data types, and the required measures are aggregated (for example, average student AI usage). Calculated fields and filters are prepared to support analysis.

Step 4 - Six Visualizations. Six worksheets are created: AI Adoption Trend Over Years, AI Adoption By Country, Average Student AI Usage, Student vs Teacher AI Usage, Rural AI Usage, and AI Usage by Top AI Tool.

Step 5 - Dashboard. The worksheets are arranged together on one dashboard so that all key indicators can be viewed on a single screen.

Step 6 - Six-Scene Story. A Tableau Story is created with an Executive Overview scene followed by one scene each for the trend, countries, average student usage, student vs teacher usage and AI tools.

Step 7 - Tableau Public. The workbook is published to Tableau Public so that the dashboard and story are available online and can be embedded.

Step 8 - Flask Web Application. A Flask application with HTML, CSS and JavaScript embeds the published views and offers separate buttons for the Dashboard and the Story.

5. Dataset Description

Item	Details
Dataset name	Global AI in Education (CSV file)
Number of rows	[Enter row count from your CSV]
Number of columns	12 (3 categorical + 9 numeric)
Time period	2015 - 2026
Countries covered	Australia, Brazil, Canada, China, Germany, India, Nigeria, Pakistan, United Kingdom, United States
Nature of data	Sample / synthetic data created for learning and demonstration

The dataset represents how AI is used in education in different countries and years. Each record describes a country in a given year with usage percentages for students, teachers, urban and rural areas, together with supporting indicators such as internet penetration and education index.

Important columns

Column	Type	Meaning
Country	Dimension	Country to which the record belongs
Year	Dimension	Year of the record (2015 - 2026)
Top Ai Tool	Dimension	Most used AI tool: ChatGPT, Google Gemini, Khanmigo or Microsoft Copilot
Student Ai Usage Pct	Measure	Percentage of students using AI
Teacher Ai Usage Pct	Measure	Percentage of teachers using AI

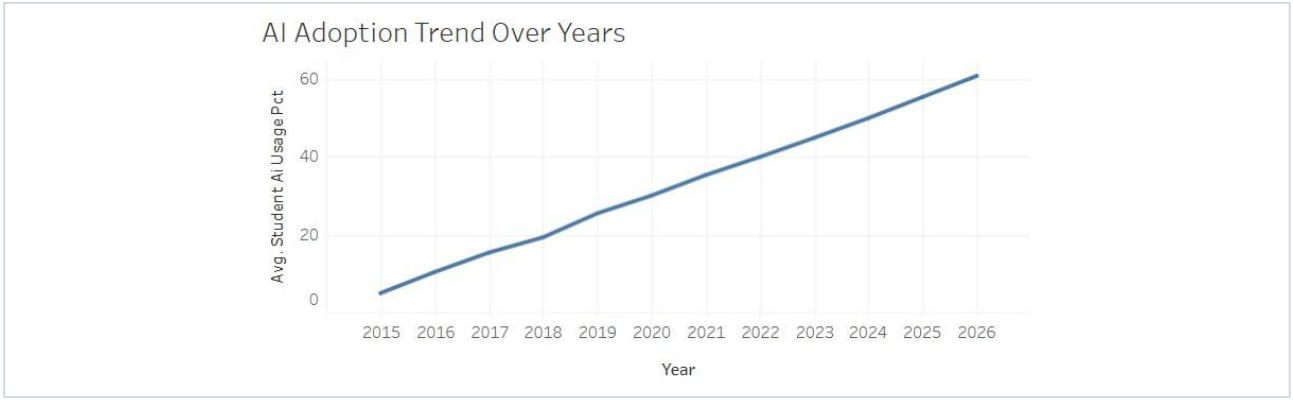
Urban Ai Usage Pct	Measure	AI usage percentage in urban areas
Rural Ai Usage Pct	Measure	AI usage percentage in rural areas
Schools Ai Adoption Pct	Measure	Percentage of schools that have adopted AI
Internet Penetration Pct	Measure	Percentage of population with internet access
Education Index	Measure	Education level indicator (0 to 1)
Gender Gap Ai Usage Pct	Measure	Gap in AI usage between genders
Avg Daily Ai Usage Min	Measure	Average minutes of AI use per day

Note: The dataset is sample / synthetic data. It is used to demonstrate visualization and storytelling skills and does not represent verified real-world statistics.

6. Tableau Visualizations

Six visualizations were created in Tableau. Each is shown below with its purpose, a description of what is displayed and an interpretation.

Figure 1: AI Adoption Trend Over Years

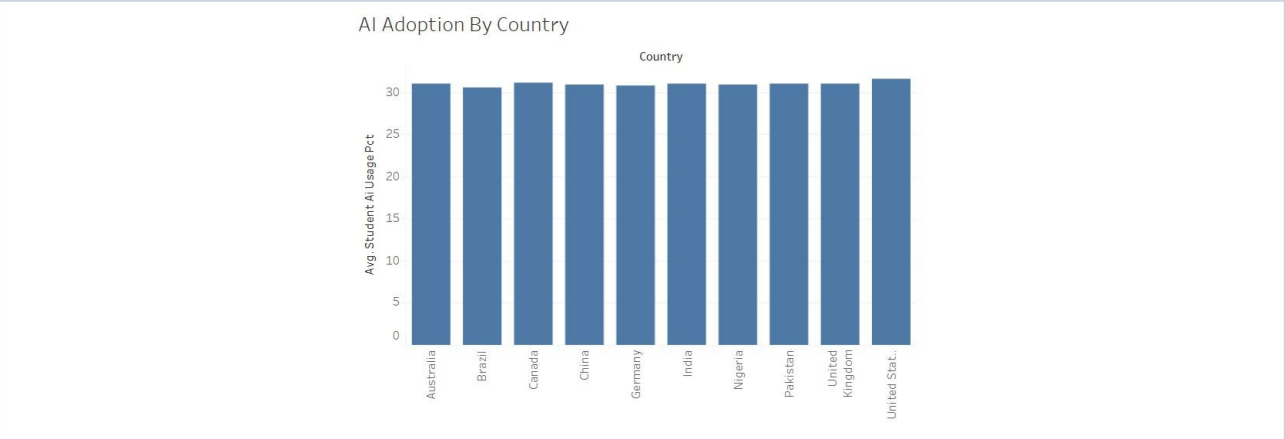


Purpose: To show how average student AI usage has changed from year to year.

Description: A line chart with Year (2015 - 2026) on the x-axis and Average Student AI Usage Pct on the y-axis.

Interpretation: Usage rises steadily and without any decline, from roughly 5% in 2015 to over 60% in 2026. Growth is fastest after 2018 and then continues at a nearly constant pace of about 5 percentage points per year, showing that AI adoption in education is expanding consistently.

Figure 2: AI Adoption By Country

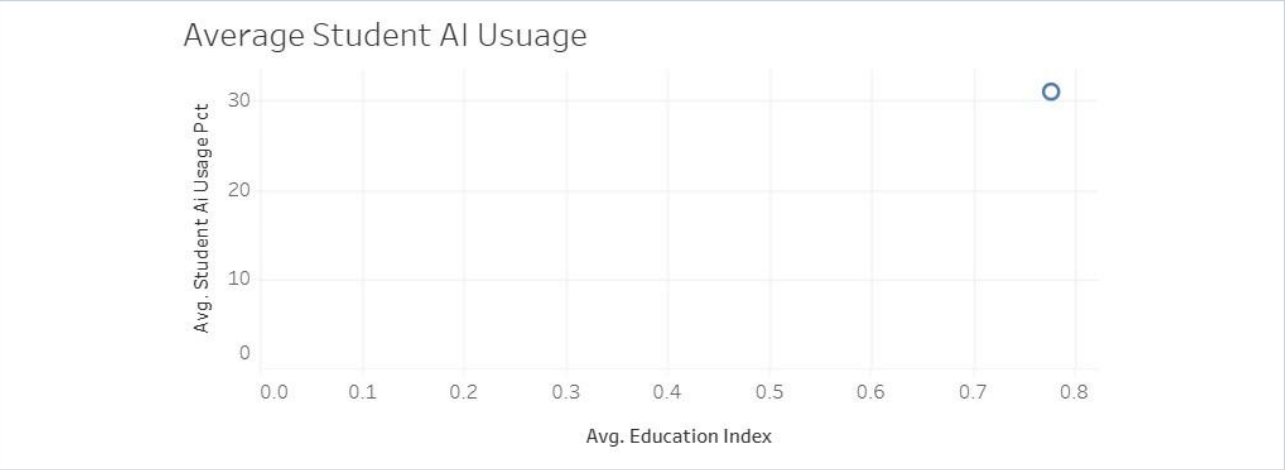


Purpose: To compare average student AI usage across the ten countries.

Description: A bar chart with one bar per country (Australia to United States) showing Average Student AI Usage Pct.

Interpretation: All ten countries have very similar averages of about 30.5% to 31.5%. The United States is marginally the highest and Brazil marginally the lowest. The small differences indicate that, in this dataset, adoption is evenly spread across countries.

Figure 3: Average Student AI Usage

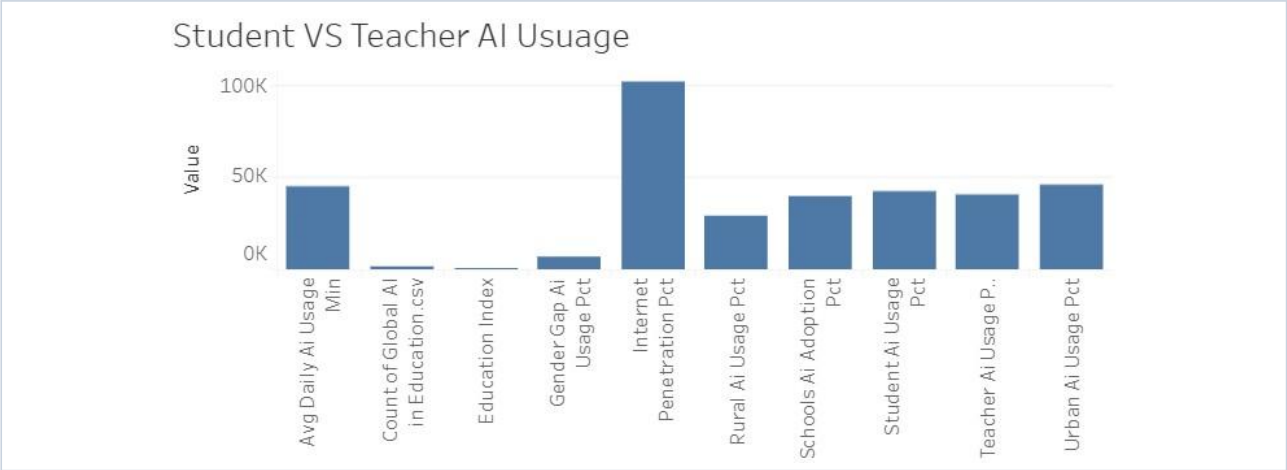


Purpose: To relate overall student AI usage to the education index.

Description: A scatter plot with Average Education Index on the x-axis and Average Student AI Usage Pct on the y-axis.

Interpretation: Because both values are averaged over the entire dataset, a single point appears, at an education index of about 0.78 and student AI usage of about 31%. It acts as a summary indicator: a fairly high education index goes together with moderate student AI usage.

Figure 4: Student vs Teacher AI Usage

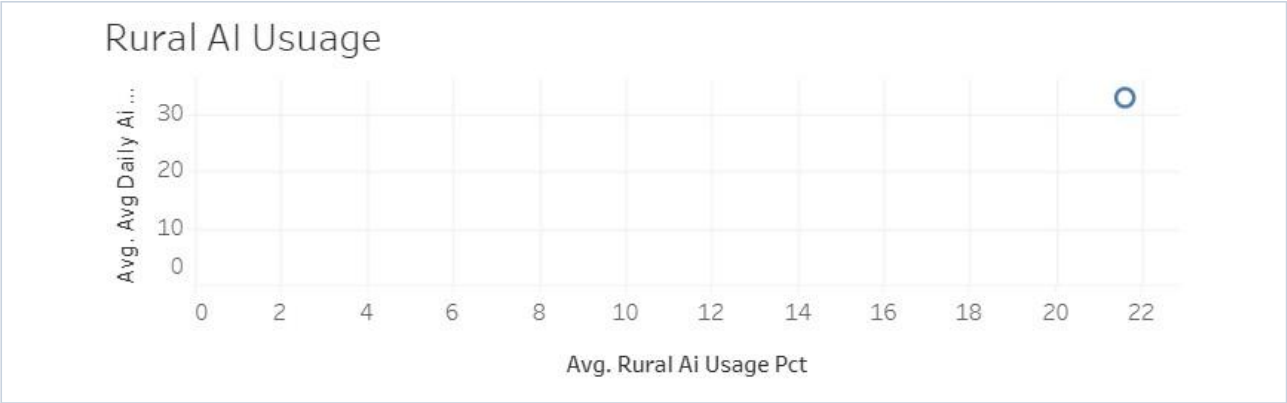


Purpose: To compare student and teacher AI usage alongside other indicators.

Description: A bar chart of measure values (summed) for Avg Daily AI Usage, Gender Gap, Internet Penetration, Rural, Urban, Schools Adoption, Student and Teacher AI Usage.

Interpretation: Student usage is slightly higher than teacher usage, and urban usage is higher than rural usage. Internet penetration has the largest bar. Since these are summed values, they are best read as relative comparisons between bars rather than as exact percentages.

Figure 5: Rural AI Usage

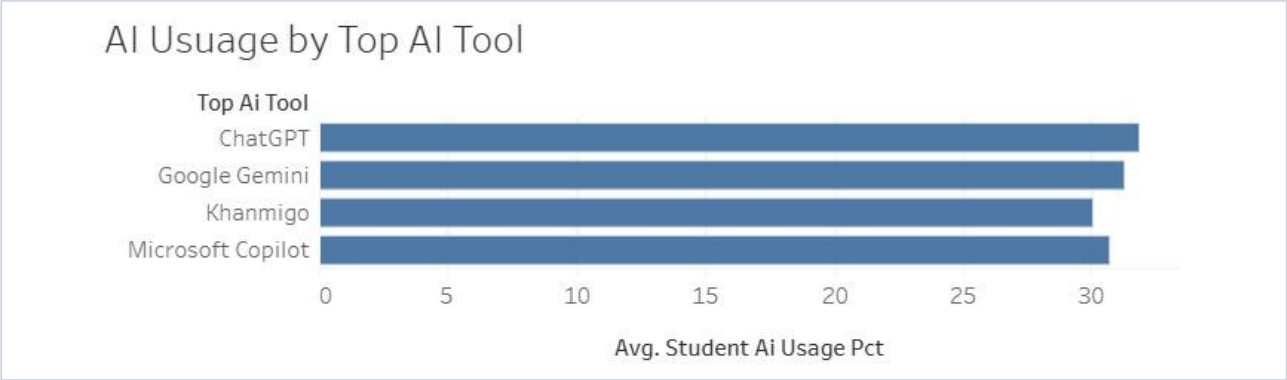


Purpose: To show rural AI usage against the average daily time spent on AI.

Description: A scatter plot with Average Rural AI Usage Pct on the x-axis and Average Daily AI Usage (minutes) on the y-axis.

Interpretation: The single point shows rural AI usage of about 21 - 22% with an average of about 30 minutes of AI use per day. Rural usage is clearly lower than the overall student average of about 31%, indicating a rural-urban gap.

Figure 6: AI Usage by Top AI Tool



Purpose: To identify which AI tools are associated with the highest student AI usage.

Description: A horizontal bar chart of Average Student AI Usage Pct for each Top AI Tool: ChatGPT, Google Gemini, Khanmigo and Microsoft Copilot.

Interpretation: ChatGPT ranks first (about 32%), followed by Google Gemini, Microsoft Copilot and Khanmigo (about 30%). The bars are close in length, so no single tool is far ahead of the others.

7. Dashboard

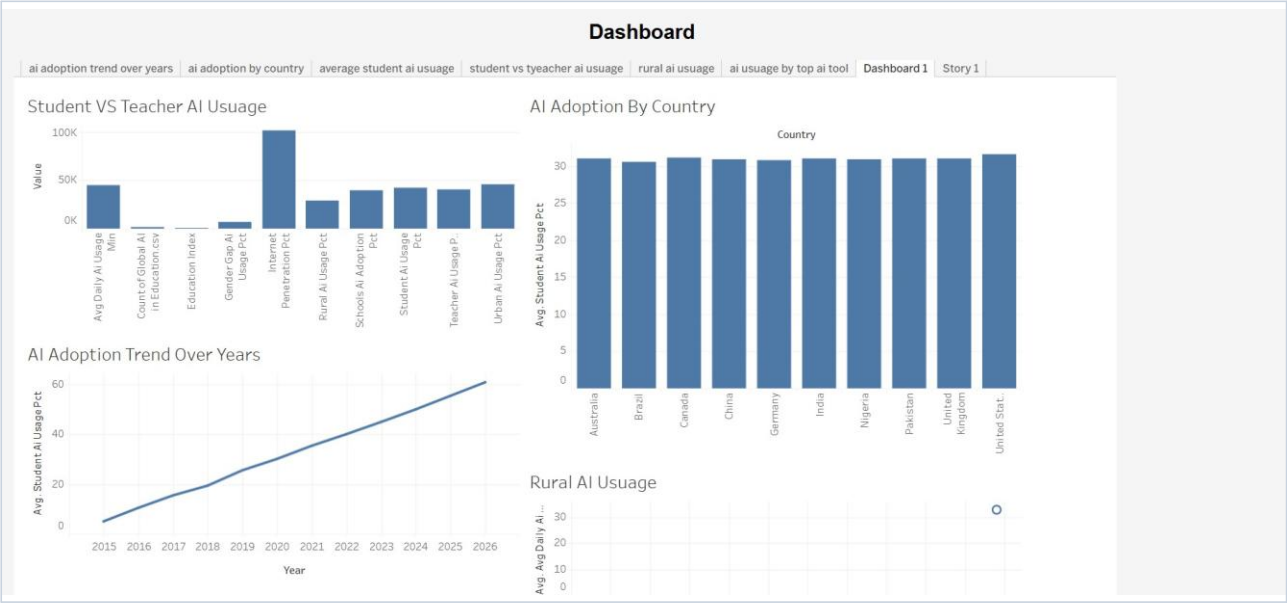


Figure 7: Tableau Dashboard (top section)

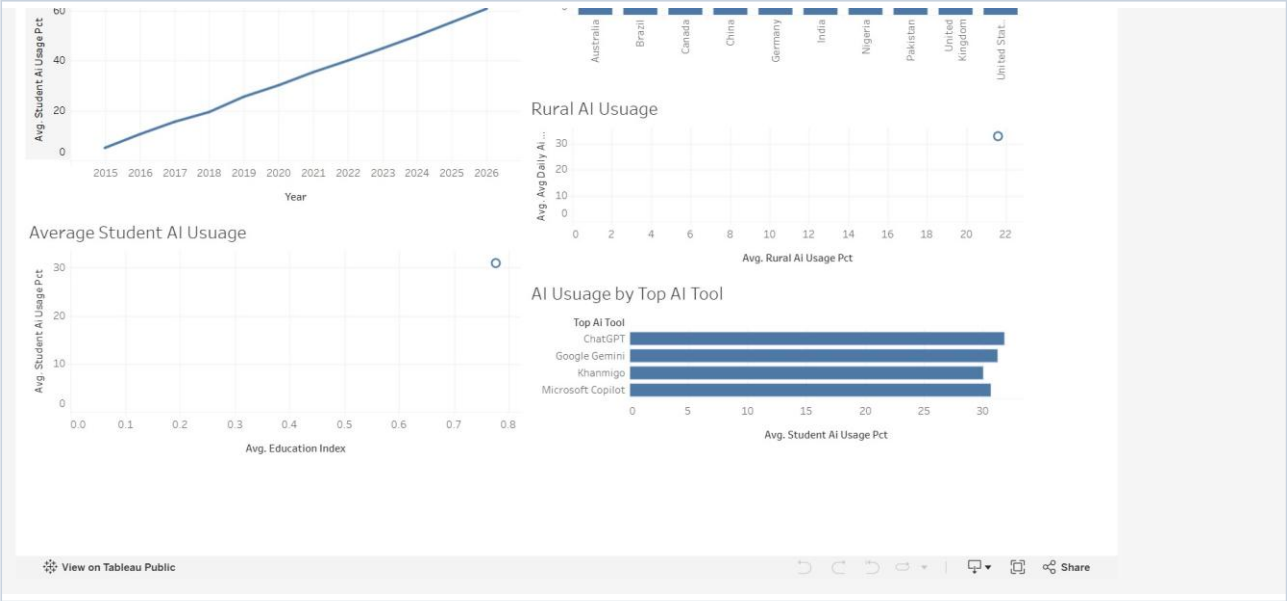


Figure 8: Tableau Dashboard (lower section, published on Tableau Public)

The Tableau Dashboard combines the six visualizations into a single interactive interface. It allows users to explore AI adoption, usage, tools and regional differences in one place. Calculated fields such as Adoption Category and Impact Category were also used to support the analysis.

Filters and interaction

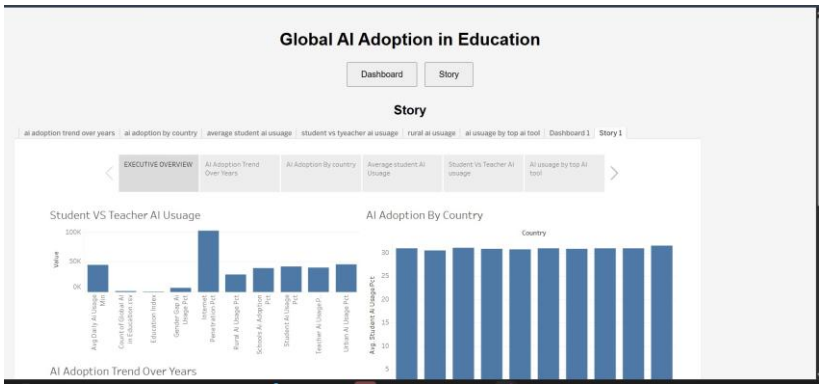
- **Country filter:** restricts all charts to one or more selected countries.
- **Year filter:** limits the data to a chosen period so that trends can be compared across years.
- **Top AI Tool filter:** shows results for a selected AI tool only.

- Hovering over any mark displays a tooltip with its exact value.

8. Tableau Story

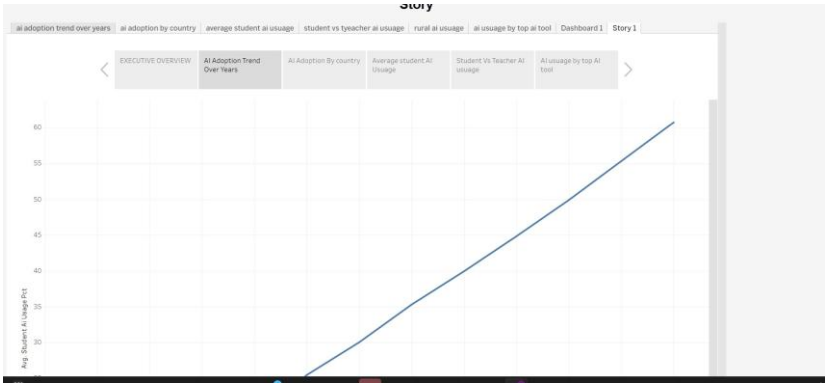
The Tableau Story arranges the analysis as a sequence of six scenes, moving from an overview to the individual findings.

Figure 9: Scene 1 - Executive Overview



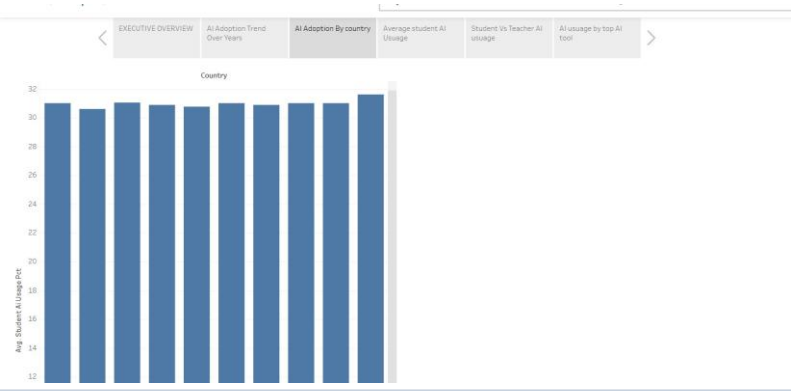
Explanation: Gives the complete picture at a glance, combining the student vs teacher comparison, the country comparison, the adoption trend and the other key charts.

Figure 10: Scene 2 - AI Adoption Trend Over Years



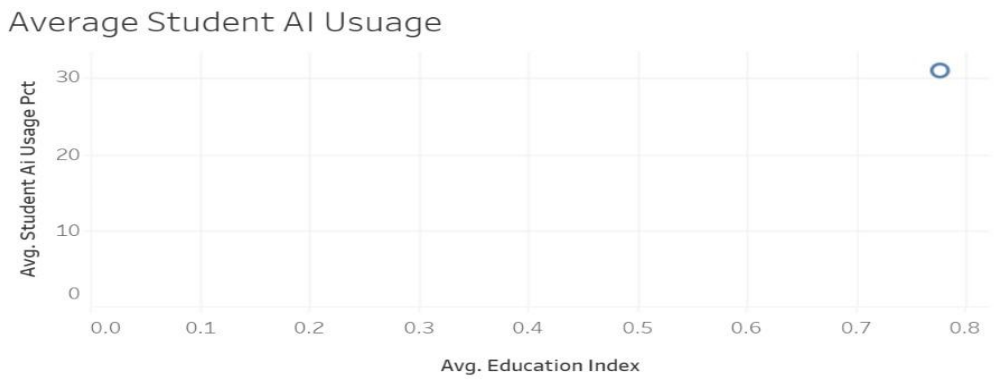
Explanation: Shows that average student AI usage grows steadily from about 5% (2015) to over 60% (2026).

Figure 11: Scene 3 - AI Adoption By Country



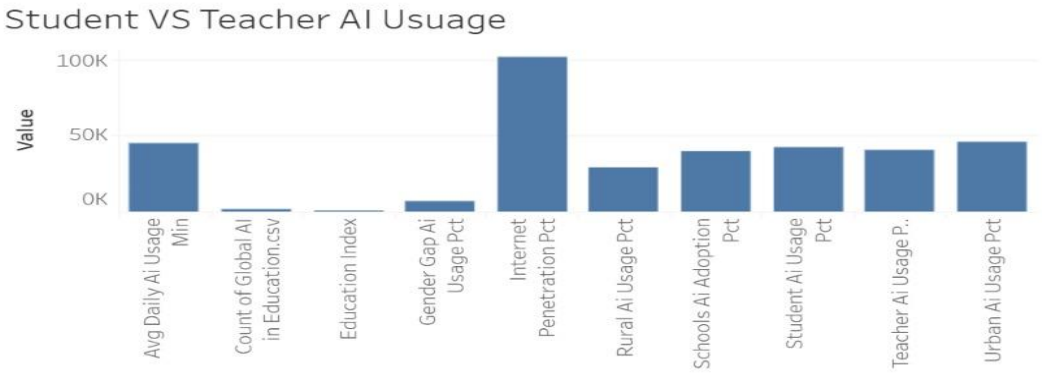
Explanation: Compares the ten countries; all stay close to 31%, with the United States slightly ahead.

Figure 12: Scene 4 - Average Student AI Usage



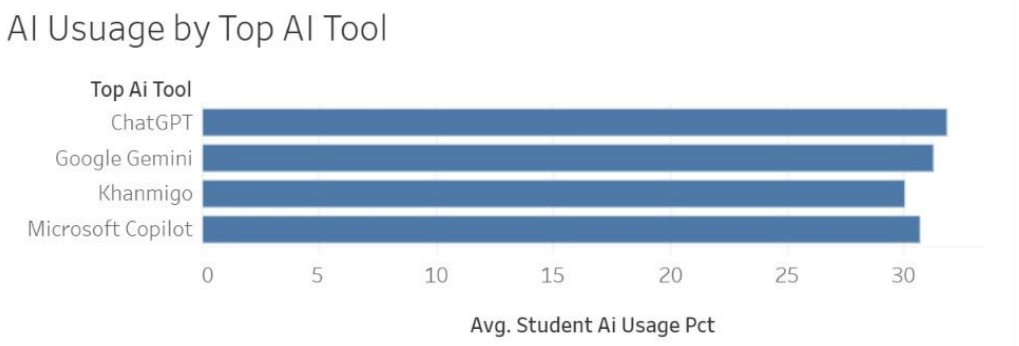
Explanation: Shows overall student AI usage (about 31%) against the average education index (about 0.78).

Figure 13: Scene 5 - Student vs Teacher AI Usage



Explanation: Compares student and teacher usage with the other indicators; student usage is slightly higher than teacher usage.

Figure 14: Scene 6 - AI Usage by Top AI Tool



Explanation: Ranks the four AI tools: ChatGPT leads, followed by Google Gemini, Microsoft Copilot and Khanmigo.

9. Flask Web Integration

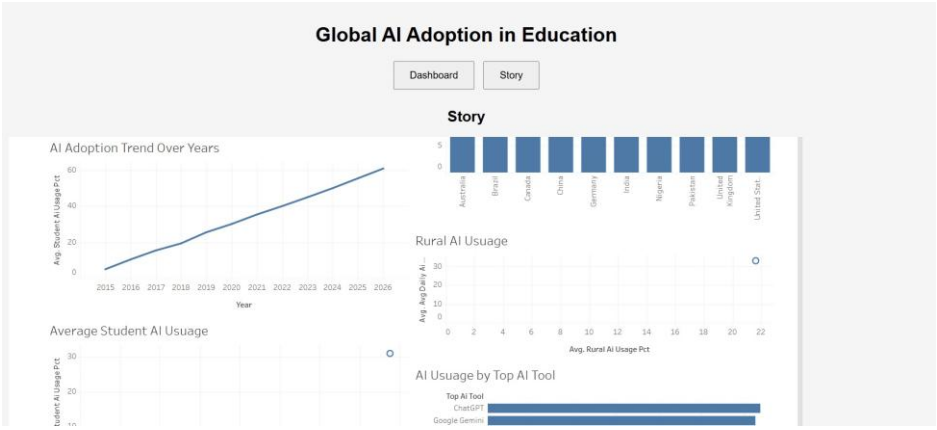


Figure 15: Flask web page with Dashboard and Story buttons

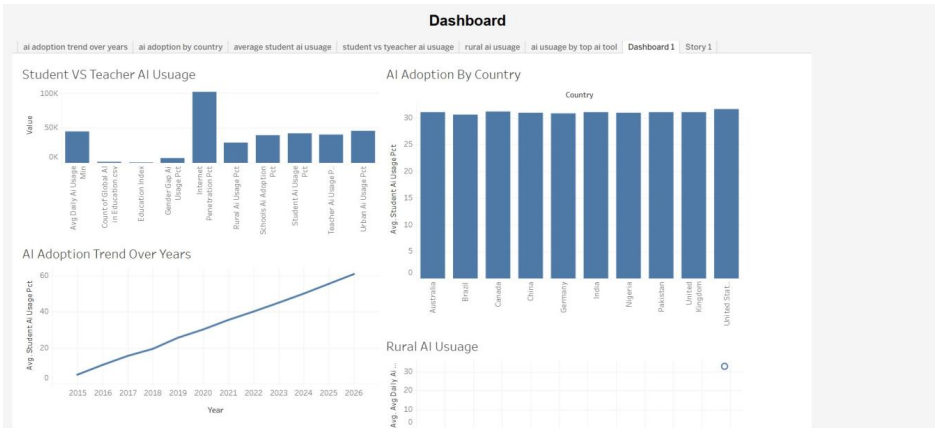


Figure 16: Tableau Dashboard displayed inside the Flask application



Figure 17: Tableau Story displayed inside the Flask application

A Flask-based web application was developed to integrate the Tableau Dashboard and Story. The published Tableau views were embedded into the Flask application using HTML, CSS and JavaScript. The application provides separate buttons for accessing the Dashboard and the Story.

10. Results / Key Insights

- **Steady growth:** average student AI usage rose from about 5% in 2015 to over 60% in 2026 with no year of decline.
- **Balanced across countries:** all ten countries record similar averages (about 30.5% - 31.5%); the United States is slightly highest.
- **Students vs teachers:** students use AI slightly more than teachers.
- **Urban vs rural:** urban usage is higher than rural usage; rural AI usage is only about 21 - 22%, so a rural gap exists.
- **Tools:** ChatGPT is the leading tool, followed closely by Google Gemini, Microsoft Copilot and Khanmigo.
- **Context:** the overall education index is about 0.78 and average AI use is about 30 minutes per day.

11. Limitations

The dataset used in this project is sample / synthetic data. The findings are therefore meant for demonstration and analysis of visualization techniques and should not be treated as verified global statistics. Because the values are averaged, several charts (such as the country comparison) show very small differences, and some scatter plots display only a single summary point. Using more detailed, real-world data would give richer comparisons.

12. Conclusion

This project presented the story of AI adoption in education through Tableau. Six visualizations were designed, combined into one dashboard, and organised into a six-scene story that moves from an overview to trend, country, usage and tool comparisons. The work was published on Tableau Public and embedded in a Flask web application with separate Dashboard and Story pages.

The analysis shows that AI usage among students has grown steadily, that students use AI slightly more than teachers, that rural usage lags behind urban usage, and that ChatGPT is the most used tool. Although the data is synthetic, the project demonstrates a complete workflow from data to dashboard, story and web deployment.

13. Future Scope

- Replace the sample dataset with real data from surveys and official education sources.
- Add more indicators such as the impact of AI on learning outcomes and the barriers to adoption (cost, training, connectivity).
- Add state / city level maps to show geographical differences in AI adoption.
- Add forecasting to predict AI adoption in the coming years.
- Connect the Flask application directly to a database for automatic updates.
- Add user login and download options for reports from the web application.

- Deploy the Flask application on a cloud platform so it can be accessed publicly.

14. References

- Tableau Public - public.tableau.com (publishing and hosting the dashboard and story).
- Tableau Desktop / Tableau Public documentation - help.tableau.com.
- Flask documentation - flask.palletsprojects.com.
- HTML, CSS and JavaScript documentation - developer.mozilla.org.
- Dataset: *Global AI in Education* (CSV), sample / synthetic data.
- Published project: "Global ai Adoption in Education Final" by Yashraj Gadilkar , Tableau Public.